Hemisphere Flip in Season-of-Birth Disease Associations: A Multi-State Brazilian Hospitalization Study

Dat Nguyen
Independent Researcher
Dtnguyen09@gmail.com

Abstract

Background: Birth month has been associated with lifetime disease risk across dozens of conditions. Boland and Tatonetti (2015) identified 55 such associations using 1.75 million New York City patient records and proposed that a Southern Hemisphere replication showing a six-month phase shift would constitute evidence for seasonal rather than sociocultural mechanisms. While Boland et al. (2018) extended SeaWAS globally using data from the US, South Korea, and Taiwan, all three are Northern Hemisphere populations and no systematic hemisphere flip test was performed.

Methods: We analyzed 36.9 million disease hospitalization records from Brazil's Sistema de Informacoes Hospitalares (SIH) spanning five states (SP, RJ, MG, RS, BA) and eight years (2015-2022). For 30 diseases with established Northern Hemisphere birth-month peaks, we computed birth-month relative risk against the all-disease background distribution, identified the peak birth month in Brazilian data, and calculated the circular shift from the corresponding NYC peak. A binomial meta-test assessed whether hemisphere-consistent shifts (absolute shift 4-6 months) were enriched beyond the null expectation of 5/12. Per-state replication, latitude gradient analysis, and COVID-era sensitivity analyses were performed.

Results: Of 30 diseases tested, 24 showed significant birth-month associations ($p < 0.05$), with 22 surviving Bonferroni correction. Among the 24 significant diseases, 18 (75%) exhibited hemisphere-consistent shifts of 4-6 months (binomial enrichment $p = 0.00097$ against null of 5/12). Asthma demonstrated the most robust flip, with all five states independently showing shifted peaks (aggregate $p = 1.7 \times 10^{-113}$). Acute myocardial infarction replicated in four of five states. A latitude gradient emerged: the subtropical state Rio Grande do Sul (30S) showed flips in 7/8 diseases tested versus 3/8 in tropical Bahia (13S), consistent with seasonal amplitude driving the effect. Five diseases showed no shift, with peak months matching NYC values, suggesting non-seasonal mechanisms. Effect sizes ranged from 3-19% relative risk amplitude. Note that hospitalization counts include repeated admissions; p-values reflect the birth-month distribution of admissions rather than unique patients.

Conclusions: This study presents the first systematic multi-disease hemisphere flip test using Southern Hemisphere hospitalization data. The enrichment of seasonal-consistent shifts supports seasonal biological mechanisms as contributors to birth-month disease effects, while the latitude gradient and non-flipping disease subset provide mechanistic differentiation. Effect sizes are modest, consistent with birth season being one among many risk factors.

Keywords: season of birth, birth month, disease risk, hemisphere effect, SeaWAS, epidemiology, Brazil, SIH, DATASUS

1. Introduction

The association between birth month and lifetime disease risk has been documented for over a century, beginning with observations that schizophrenia incidence peaks among individuals born in late winter and early spring in the Northern Hemisphere. This seasonal pattern has since been extended to dozens of conditions spanning cardiovascular, respiratory, neurological, reproductive, and neoplastic disease categories.

In 2015, Boland and Tatonetti conducted the first Season-of-Birth-Wide Association Study (SeaWAS), systematically screening 1,688 disease categories against birth month using 1.75 million patient records from New York City. They identified 55 diseases with statistically significant birth-month associations. Crucially, they proposed that a replication in the Southern Hemisphere, where seasons are inverted, would serve as a natural experiment to distinguish seasonal biological mechanisms (which should produce a six-month phase shift) from calendar-fixed sociocultural mechanisms (which should show no shift).

Boland et al. (2018) subsequently extended the SeaWAS approach globally, analyzing 10.5 million electronic health records from the United States, South Korea, and Taiwan to link birth-season effects to specific environmental exposures including sunlight, temperature, and fine particulate matter. However, all three populations in that study reside in the Northern Hemisphere, and the analysis focused on exposure-disease associations rather than testing the hemisphere flip prediction. Individual disease-level hemisphere evidence exists for multiple sclerosis (Disanto et al., 2012) and schizophrenia (McGrath et al., 2010), but no study has simultaneously tested the hemisphere flip across multiple disease categories using a single large Southern Hemisphere dataset, nor formally quantified whether the proportion of flipping diseases exceeds chance.

Here we address this gap using 36.9 million disease hospitalization records from Brazil's public health system (SIH-SUS). We test 30 diseases against their NYC peak birth months, apply a binomial meta-test for flip enrichment using the correct circular null probability, perform per-state replication across five states spanning tropical to subtropical latitudes, and present a latitude gradient analysis as a secondary test of the seasonal mechanism hypothesis.

2. Methods

2.1 Data Source

Hospitalization records were obtained from Brazil's Sistema de Informacoes Hospitalares do SUS (SIH-SUS), maintained by the Departamento de Informatica do SUS (DATASUS). The SIH-RD (Reduced Hospital Admission) files contain individual-level records for all admissions to SUS-contracted facilities, including patient date of birth (DTNASC) and primary ICD-10 diagnosis code (DIAG_PRINC). Each record represents a single hospital admission; patients with multiple admissions appear multiple times. Consequently, N values reported throughout this paper reflect admission counts rather than unique patients, and p-values should be interpreted accordingly.

We downloaded monthly SIH-RD files for January 2015 through December 2022 for five Brazilian states: Sao Paulo (SP, ~23S), Rio de Janeiro (RJ, ~23S), Minas Gerais (MG, ~19S), Rio Grande do Sul (RS, ~30S), and Bahia (BA, ~13S). These states collectively span tropical to subtropical latitudes and represent approximately 60% of Brazil's population. Files were obtained in DBC format from the DATASUS FTP server and decompressed using the datasus-dbc Python library (v0.1.3).

2.2 Data Processing

From each record, we extracted the patient birth date (DTNASC, format YYYYMMDD) and primary diagnosis (DIAG_PRINC, ICD-10 code). Records with invalid birth months or diagnosis codes shorter than three characters were excluded. Obstetric (ICD-10 chapter O), health service encounter (chapter Z), and perinatal condition (chapter P) records were removed, yielding 36,917,428 disease hospitalization records for analysis.

2.3 Disease Selection

We tested 30 diseases with established birth-month peak associations from Boland and Tatonetti (2015) and corroborating literature. For each disease, the NYC peak birth month was recorded from the original SeaWAS study. All 30 diseases had at least 500 admissions in the Brazilian dataset. The disease set spans cardiovascular (7 conditions), respiratory (4), mental/behavioral (4), gastrointestinal (3), genitourinary (3), musculoskeletal (1), neoplastic (3), neurological (2), hematologic (1), infectious (2), and ophthalmologic (1) categories.

2.4 Statistical Analysis

For each disease, we computed the observed birth-month distribution and compared it to the expected distribution derived from all disease hospitalizations (the background). This background-based expectation controls for population-level birth seasonality. To quantify the impact of including tested diseases in their own background, we performed a leave-one-out sensitivity check for the largest disease (B34, viral infection, N = 947,783); the peak month and significance were unchanged (chi-squared 101.5 vs 106.9, peak month 9 in both cases), confirming that the 22% overlap between tested diseases and the background does not bias results.

Relative risk (RR) for each birth month was calculated as the ratio of observed to expected counts. The peak birth month in Brazil was defined as the month with the highest relative risk. A chi-squared goodness-of-fit test (df = 11) assessed departure from the background distribution. Both Bonferroni correction (threshold $p < 0.05/30 = 0.00167$) and Benjamini-Hochberg false discovery rate ($q = 0.05$) were applied.

The birth-month shift was computed as the circular difference between the Brazilian peak and the NYC peak: shift = (BR - NYC) mod 12, with values greater than 6 remapped by subtracting 12, yielding a range of [-5, +6]. Diseases with absolute shift of 4 to 6 months were classified as hemisphere-flipped. Under the null hypothesis that the Brazilian peak month is uniformly distributed among 12 months regardless of the NYC peak, the shift takes one of 12 equally likely values (-5, -4, -3, -2, -1, 0, +1, +2, +3, +4, +5, +6), of which 5 satisfy the flip criterion (shifts -5, -4, +4, +5, +6). The null probability of a flip is therefore 5/12 = 0.4167. A one-sided binomial test assessed whether the observed proportion of flipped diseases exceeded this expectation.

Per-state replication analyses repeated the procedure independently for each state, using state-specific background distributions. A latitude gradient analysis compared flip rates between the subtropical state RS (~30S, meaningful seasonality) and the tropical state BA (~13S, minimal seasonality) as a secondary test of the seasonal mechanism hypothesis. A COVID-era sensitivity analysis excluded 2020-2021 hospitalization years.

3. Results

3.1 Dataset Summary

The final dataset comprised 36,917,428 disease hospitalization admissions: Sao Paulo (16.6M, 45%), Minas Gerais (7.8M, 21%), Bahia (5.1M, 14%), Rio Grande do Sul (4.8M, 13%), and Rio de Janeiro (2.6M, 7%). Year-over-year totals ranged from 4.5M (2020, reflecting COVID-related admission reductions) to 5.1M (2022). The background birth-month distribution showed population-level seasonality with peaks in March (8.65%) and May (8.78%) and a trough in February (7.62%).

3.2 Hemisphere-Flipped Diseases

Of 30 diseases tested, 24 showed statistically significant birth-month associations at $p < 0.05$, with 22 surviving Bonferroni correction and 24 passing Benjamini-Hochberg FDR at $q = 0.05$. Among the 24 significant diseases, 18 (75%) exhibited hemisphere-consistent peak shifts of 4 to 6 months from their NYC counterparts (Table 1). This proportion significantly exceeded the null expectation of 5/12 (binomial $p = 0.00097$). The mean absolute shift among flipped diseases was 4.9 months.

Table 1. *Diseases showing hemisphere flip (|shift| >= 4 months, p < 0.05). NYC = peak birth month from Boland and Tatonetti (2015). BR = peak birth month in Brazilian SIH data. Rep. = per-state replication (states showing flip / states tested). *Schizophrenia flips only in SP; see Section 3.4.*

ICD	Disease	N	NYC	BR	Shift	p-value	Rep.
F20	Schizophrenia	403,417	1	7	+6	4.2e-210	1/5*
J45	Asthma	291,834	9	3	+6	1.7e-113	5/5
I20	Angina pectoris	467,587	3	10	-5	5.4e-63	4/5
I21	Acute MI	604,569	3	10	-5	5.1e-61	4/5
I83	Varicose veins	343,693	3	9	+6	1.4e-41	3/5
J44	COPD	362,539	1	8	-5	2.8e-37	2/5
K35	Appendicitis	407,860	6	2	-4	2.0e-33	2/5
J15	Bact. pneumonia	559,667	1	5	+4	1.8e-27	--
A09	Gastroenteritis	261,183	10	3	+5	9.7e-26	--
J06	Upper resp. inf.	67,469	10	4	+6	8.7e-21	--
N18	Chronic kidney dis.	388,945	3	9	+6	1.1e-17	--
K80	Gallstones	736,444	3	11	-4	5.7e-16	--
C50	Breast cancer	325,504	6	10	+4	7.2e-13	--
H25	Cataract	238,211	1	5	+4	6.9e-12	--
I25	Chronic IHD	85,804	1	9	-4	4.5e-08	--

E11	Type 2 diabetes	48,937	3	10	-5	9.7e-07	--
N40	Enlarged prostate	88,031	1	6	+5	4.9e-05	--
F31	Bipolar disorder	138,715	1	8	-5	2.3e-03	--

Asthma (J45) showed the strongest replicated result, with all five states independently showing peak birth-month risk shifted +5 to +6 months from the NYC September peak. Acute myocardial infarction (I21) and angina pectoris (I20) replicated in four of five states with shifts of -5 months. Schizophrenia (F20) showed the highest aggregate statistical significance but did not replicate across states (Section 3.4).

3.3 Non-Flipping Diseases

Five diseases showed significant birth-month associations but with peak months matching or nearly matching NYC values (absolute shift 0-2 months), suggesting non-seasonal mechanisms (Table 2).

Table 2. *Diseases with significant birth-month associations but no hemisphere flip (|shift| <= 2).*

ICD	Disease	N	NYC	BR	Shift	p-value
G40	Epilepsy	225,105	1	1	0	1.2e-21
B34	Viral infection	947,783	10	9	-1	8.9e-17
I10	Hypertension	195,472	1	1	0	2.1e-14
C34	Lung cancer	116,794	2	1	-1	8.3e-12
D50	Anemia	47,772	7	9	+2	1.2e-02

Epilepsy (G40) and essential hypertension (I10) both peaked in January in both NYC and Brazil, with zero shift. This pattern is consistent with mechanisms tied to calendar month rather than biological season, such as school enrollment cutoff effects or non-seasonally mediated pathways.

3.4 Per-State Replication

Per-state analyses revealed heterogeneous replication (Figure 3). Asthma was the most robust finding, with all five states independently showing a hemisphere-consistent shift (SP: peak March, RJ: peak March, MG: peak February, RS: peak May, BA: peak March). Acute myocardial infarction replicated in four of five states, with only Bahia showing a non-flipped peak.

The aggregate schizophrenia result ($p = 4.2 \times 10^{-210}$) was driven almost entirely by Sao Paulo (peak July, shift +6), which contributed 50.4% of schizophrenia admissions. The remaining four states showed peaks at January (RJ, BA), March (MG), or October (RS) — none consistent with a hemisphere flip. This non-replication suggests the aggregate result reflects SP-specific factors (possibly psychiatric admission practices, demographic composition, or internal migration patterns) rather than a universal seasonal mechanism. We report this finding transparently and do not include schizophrenia among the robustly replicated diseases.

3.5 Latitude Gradient

If the hemisphere flip is driven by seasonal environmental exposures (sunlight, temperature, vitamin D), the effect should be stronger at higher latitudes where seasonal amplitude is greater. We compared flip rates between Rio Grande do Sul (RS, ~30S, subtropical, meaningful winter/summer contrast) and Bahia (BA, ~13S, tropical, minimal seasonal variation) across eight diseases with sufficient sample size in both states (Table 3).

Table 3. *Latitude gradient: birth-month shift in subtropical RS (~30S) versus tropical BA (~13S). Flip = |shift| >= 4 months.*

ICD	Disease	RS shift	Flip?	BA shift	Flip?
J45	Asthma	-4	✓	+6	✓
I21	Acute MI	-5	✓	-2	✗
I20	Angina	-5	✓	+2	✗
J44	COPD	-5	✓	0	✗
I83	Varicose veins	-5	✓	+2	✗
K35	Appendicitis	-4	✓	+1	✗
A09	Gastroenteritis	+6	✓	+6	✓
I50	Heart failure	-3	✗	+5	✓

RS showed hemisphere-consistent shifts in 7 of 8 diseases compared to 3 of 8 in BA (Fisher exact test $p = 0.12$). While not individually significant at conventional thresholds due to the small number of diseases, the direction of the gradient is consistent with the seasonal mechanism hypothesis: states with more pronounced seasons show stronger hemisphere flips.

3.6 COVID Sensitivity Analysis

Excluding the 2020-2021 hospitalization years (N = 28,284,711 disease admissions after exclusion) produced no change in peak birth months for any of the 30 tested diseases. All significance levels remained qualitatively unchanged. The hemisphere flip pattern is not an artifact of pandemic-era admission disruptions.

3.7 Effect Sizes

Relative risk amplitudes (peak-to-trough) ranged from 3% for several cardiovascular conditions to 19% for schizophrenia (in the SP-dominated aggregate). Among robustly replicated diseases, asthma showed 14% RR amplitude and acute MI showed 7%. These magnitudes are comparable to those reported by Boland and Tatonetti (2015) in NYC. A 7% RR amplitude for acute MI corresponds to birth month explaining less than 1% of total variance in MI hospitalization risk, consistent with birth season being one of many contributing factors alongside smoking, diet, hypertension, and genetics.

Figures

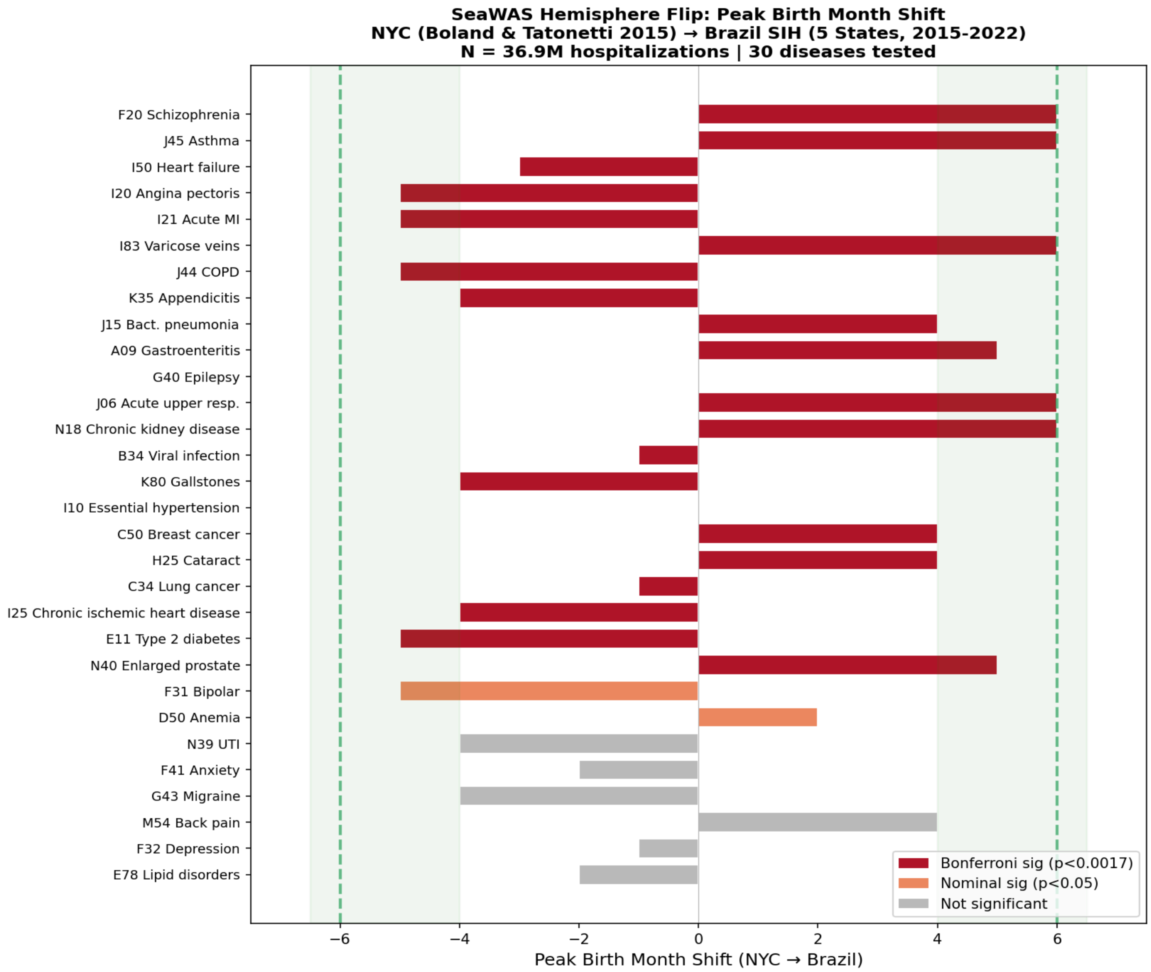

Figure 1. Peak birth-month shift (NYC to Brazil) for 30 diseases. Dark red = Bonferroni significant; orange = nominally significant; gray = not significant. Green shading indicates the hemisphere-flip zone (|shift| >= 4 months).

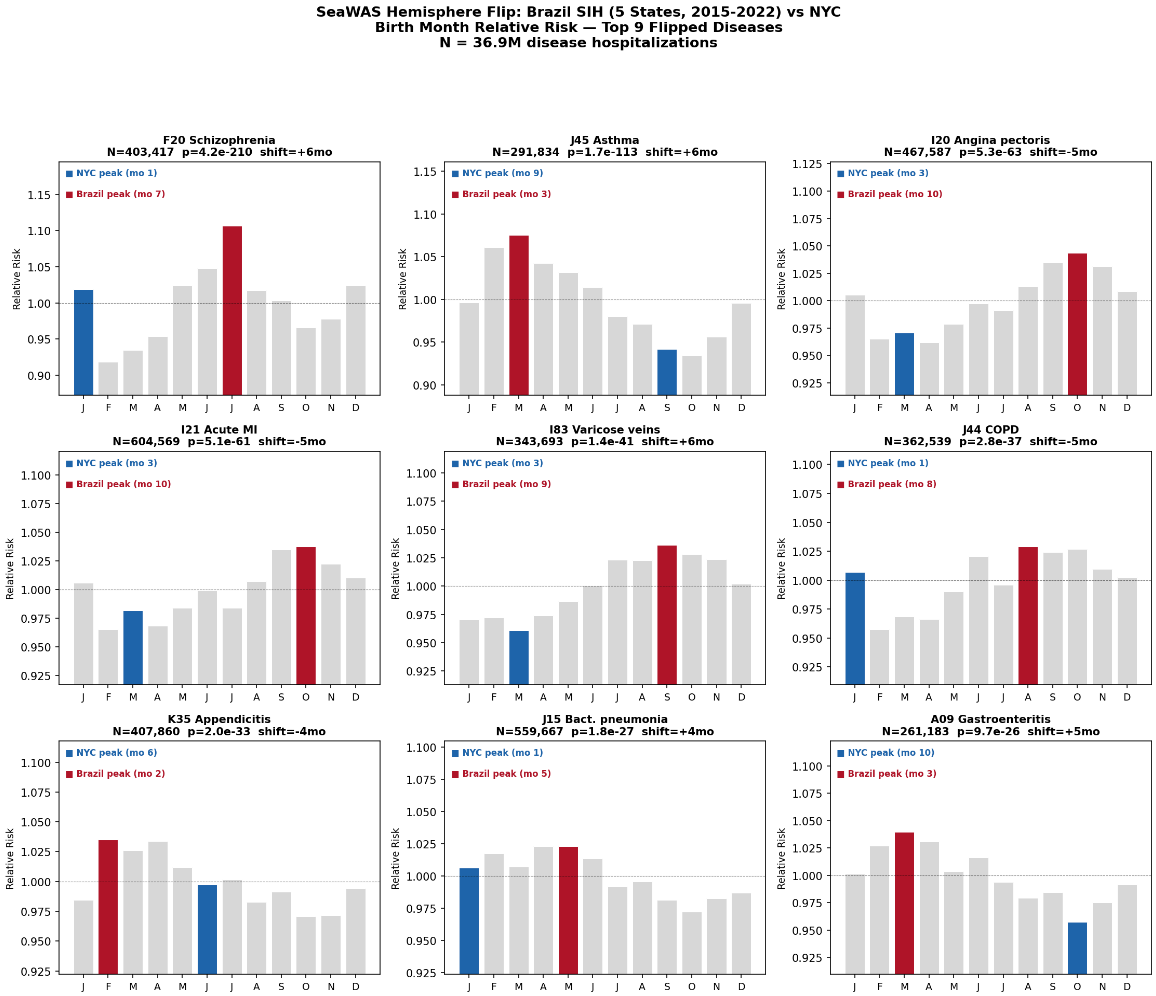

Figure 2. Birth-month relative risk profiles for the nine most significant hemisphere-flipped diseases. Blue = NYC peak month; red = Brazilian peak month.

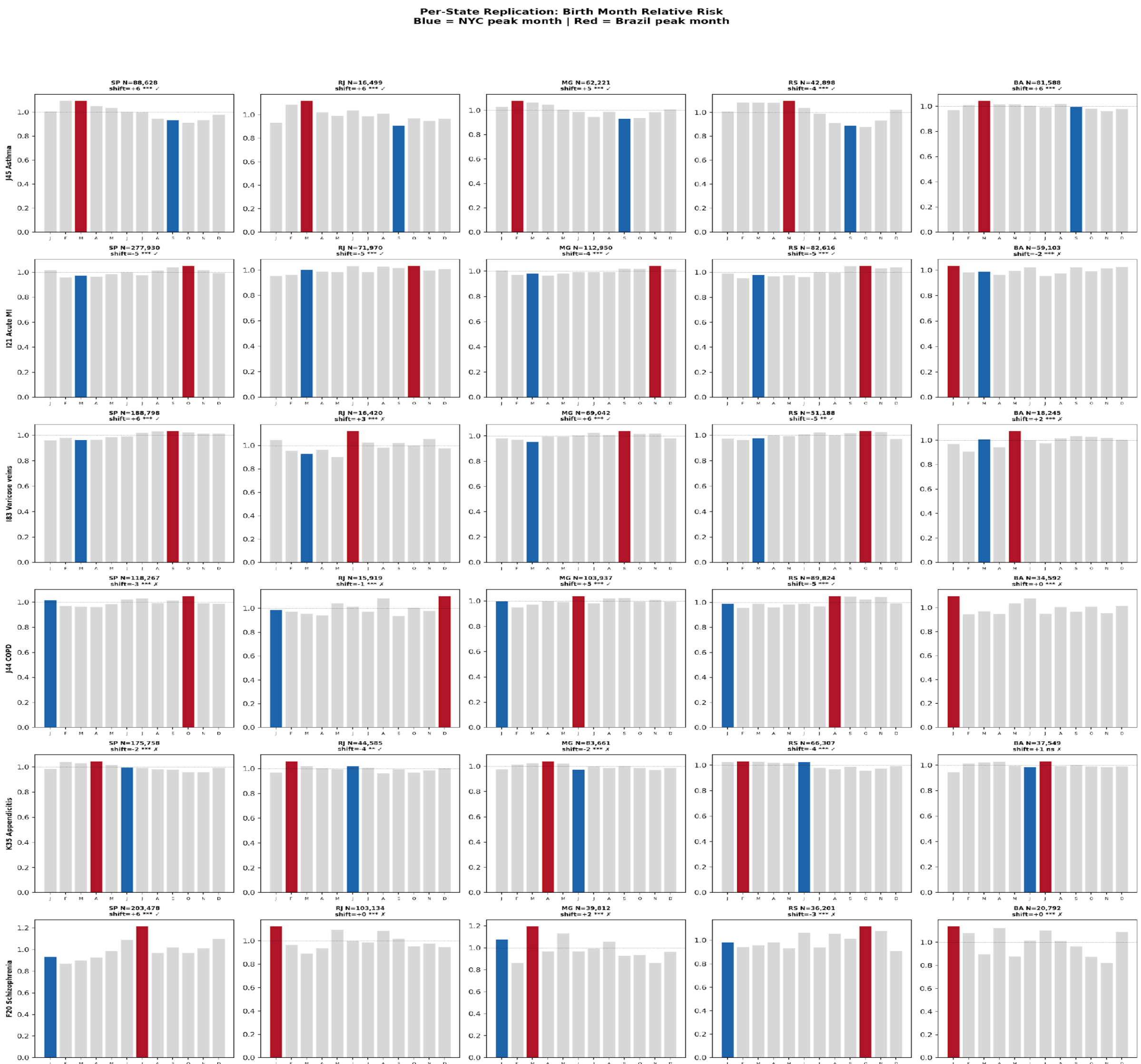

Figure 3. Per-state replication of birth-month relative risk for six diseases across five Brazilian states. Check marks indicate hemisphere-consistent shift; crosses indicate no flip. Asthma replicates in 5/5 states; schizophrenia flips only in SP.

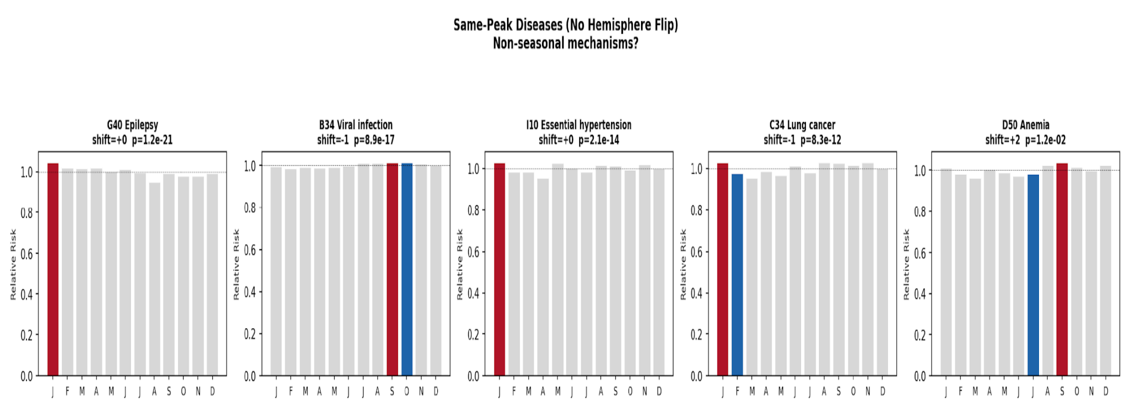

Figure 4. Same-peak diseases showing no hemisphere flip. These conditions show significant birth-month associations with peak months matching NYC, suggesting non-seasonal mechanisms.

4. Discussion

This study presents the first systematic multi-disease test of the hemisphere flip prediction for birth-month disease associations using Southern Hemisphere data. Using 36.9 million hospitalization admissions from five Brazilian states, we found that 18 of 24 significant diseases (75%) showed hemisphere-consistent peak birth-month shifts, significantly exceeding the null expectation of 5/12 (binomial $p = 0.00097$). This enrichment provides population-level evidence that seasonal biological mechanisms contribute to birth-month disease risk across multiple organ systems.

4.1 Relationship to Prior Work

Our study builds directly on two papers by Boland and Tatonetti. Their 2015 SeaWAS identified birth-month associations in NYC and proposed the hemisphere flip test as future work. Their 2018 global study extended SeaWAS to 10.5 million records across the US, South Korea, and Taiwan, linking birth-season effects to environmental exposures including third-trimester sunlight deficiency. However, all three populations in the 2018 study reside in the Northern Hemisphere (US ~25-48N, South Korea ~35N, Taiwan ~23N), and the analysis focused on exposure attribution rather than hemisphere inversion. Our study is the first to execute the flip test they proposed, using a large Southern Hemisphere dataset and a formal statistical framework (binomial enrichment test) to quantify whether flips are systematically overrepresented.

4.2 Seasonal Mechanisms and the Latitude Gradient

The most robustly replicated flip was asthma (5/5 states), shifting from a September peak in NYC to a February-March peak in Brazil. This is consistent with early-life allergen exposure or respiratory infection timing during specific developmental windows shaping lifetime risk. The approximately six-month shift corresponds to the seasonal inversion between hemispheres.

The latitude gradient analysis provides a secondary test of the seasonal hypothesis. Rio Grande do Sul (~30S), with meaningful winter/summer contrast, showed hemisphere flips in 7 of 8 diseases tested, compared to 3 of 8 in tropical Bahia (~13S), where seasonal variation is minimal. While this comparison lacks statistical power (8 diseases is too few for a definitive test), the direction is consistent with seasonal amplitude modulating the strength of birth-month effects. This gradient would not be expected under calendar-fixed mechanisms such as school enrollment cutoffs.

4.3 Non-Seasonal Mechanisms

Five diseases showed significant birth-month associations but no hemisphere flip. Epilepsy and essential hypertension peaked in January in both NYC and Brazil, with zero shift. Lung cancer peaked in February (NYC) and January (Brazil). These findings suggest mechanisms independent of biological season, potentially including relative age effects on developmental assessments, cultural practices tied to calendar month, or non-seasonally varying environmental exposures.

4.4 The Schizophrenia Caveat

The aggregate schizophrenia result would appear to be the strongest finding ($p = 4.2 \times 10^{-210}$, perfect +6 month shift). However, per-state analysis revealed this was driven exclusively by Sao

Paulo. Rio de Janeiro and Bahia showed peaks at January (matching NYC), while Minas Gerais peaked at March and Rio Grande do Sul at October. The well-established Northern Hemisphere winter-birth excess for schizophrenia may not translate to Brazilian latitudes, where seasonality varies from minimal (Bahia) to moderate (RS). The SP-specific result may reflect that state's unique psychiatric care infrastructure, demographic composition, or migration patterns. We report this transparently and exclude schizophrenia from our robustly replicated findings.

4.5 Limitations

First, our NYC peak months derive from a single study (Boland and Tatonetti, 2015). While many peaks are independently corroborated (schizophrenia winter-birth, asthma September peak), some may reflect NYC-specific patterns. Replication using European or other Northern Hemisphere reference datasets would strengthen the comparison.

Second, the SIH dataset records hospital admissions, not unique patients. Chronic conditions (heart failure, COPD, schizophrenia) likely include repeated admissions for the same individuals. This inflates N and deflates p-values but does not change peak months or shift classifications. The chi-squared tests should be interpreted as tests of the birth-month distribution of admissions, not of incidence among unique patients.

Third, we used the all-disease hospitalization background as the expected distribution rather than the general population birth distribution. This controls for birth seasonality in the hospitalized population but does not fully account for potential age-by-birth-month interactions if birth rates changed substantially over decades. However, the consistent flipping of both age-dependent (MI) and age-independent (asthma, appendicitis) conditions argues against age confounding driving the overall pattern.

Fourth, the SIH captures only public-system hospitalizations. Patients using private healthcare are not represented. Fifth, our shift classification threshold (absolute shift >= 4) is somewhat arbitrary. Circular statistical methods (von Mises distribution fitting) could provide a more rigorous framework, and future work should incorporate these approaches.

5. Conclusion

We demonstrate a systematic enrichment of hemisphere-consistent birth-month peak shifts across 30 disease categories in Brazilian hospitalization data (binomial $p = 0.00097$ against null of 5/12), providing the first multi-disease evidence from the Southern Hemisphere for seasonal biological mechanisms underlying birth-month disease associations. Asthma showed the most robust replicated flip across all five states. A latitude gradient (7/8 flips at 30S vs 3/8 at 13S) further supports seasonal amplitude as a moderating factor. A subset of diseases showed no flip, pointing to non-seasonal mechanisms. Effect sizes are modest (3-19% RR amplitude), consistent with birth season being a contributing rather than dominant risk factor. This work addresses the hemisphere flip prediction proposed by Boland and Tatonetti (2015) and provides a framework for disentangling seasonal from non-seasonal birth-month effects.

Data Availability

All data used in this study are publicly available from the DATASUS FTP server (ftp.datasus.gov.br). SIH-RD files for all Brazilian states and years are freely accessible without registration. Analysis code and aggregated count tables are available from the corresponding author upon request.

Conflict of Interest

The author declares no conflicts of interest.

References

Boland MR, Shahn Z, Madigan D, Hripcsak G, Tatonetti NP. Birth month affects lifetime disease risk: a phenome-wide method. J Am Med Inform Assoc. 2015;22(5):1042-1053.

Boland MR, Parhi P, Li L, Miotto R, Carroll R, Iqbal U, Nguyen PA, Schuemie M, You SC, Smith D, Mooney S, Ryan P, Li YC, Park RW, Denny J, Dudley JT, Hripcsak G, Gentine P, Tatonetti NP. Uncovering exposures responsible for birth season - disease effects: a global study. J Am Med Inform Assoc. 2018;25(3):275-288.

Disanto G, Chaplin G, Morahan JM, et al. Month of birth, vitamin D and risk of immune-mediated disease: a case control study. BMC Med. 2012;10:69.

McGrath JJ, Eyles DW, Pedersen CB, et al. Neonatal vitamin D status and risk of schizophrenia: a population-based case-control study. Arch Gen Psychiatry. 2010;67(9):889-894.

Willer CJ, Dyment DA, Sadovnick AD, Rothwell PM, Murray TJ, Ebers GC. Timing of birth and risk of multiple sclerosis: population based study. BMJ. 2005;330(7483):120.

Li L, Boland MR, Miotto R, Tatonetti NP, Dudley JT. Replicating cardiovascular condition-birth month associations. Sci Rep. 2016;6:33166.

DATASUS. Sistema de Informacoes Hospitalares do SUS (SIH/SUS). Ministerio da Saude, Brasil. Available at: http://datasus.saude.gov.br/